

Course 5093C

5 Credits: 4

Program Elective

Source-grounded

Chemical Engineering

To understand the basic units, unit operations & unit process in chemical engineering.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5093C is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5093C.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Polymer Technology
Course code	5093C
Course title	Chemical Engineering
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4,T:0,P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ To understand the basic units, unit operations & unit process in chemical engineering.
- □ To comprehend the mechanism of heat transfer by conduction, convection, radiation and evaporation.
- □ To explain and differentiate the mechanism of mass transfer by diffusion, absorption, leaching, extraction, adsorption and drying.
- □ To describe the basic principles and types of distillation.

Course outcomes

CO	Official outcome	Study level
CO1	14 Applying process in chemical engineering. Apply mechanism of heat transfer by conduction,	See official syllabus
CO2	convection, radiation to solve problems in chemical 15 Applying engineering. Paraphrase mechanism of mass transfer by diffusion,	See official syllabus
CO3	absorption, leaching, extraction, adsorption and 14 Understanding drying. Generalize the basic principles and types of	See official syllabus
CO4	15 Applying distillation	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3

Row	Extracted official mapping
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	engineering.	4	As specified
Module 2	solve problems in chemical engineering.	4	As specified
Module 3	extraction, adsorption and drying.	4	As specified
Module 4	Generalize the basic principles and types of distillation	4	As specified

MODULE 1

engineering.

This module is presented from the official syllabus scope for Chemical Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: engineering.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain the various systems of units 3 Understanding Describe unit operations and unit process in the

Explain the various systems of units 3 Understanding Describe unit operations and unit process in the is an official topic in Module 1 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the various systems of units 3 Understanding Describe unit operations and unit process in the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the various systems of units 3 understanding describe unit operations and unit process in the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the various systems of units 3 Understanding Describe unit operations and unit process in the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain the various systems of units 3 Understanding Describe unit operations and unit process in the definition/meaning . steps, application . Technical terms English- .

3 Understanding field of chemical engineering Solve problems related to atomic weight,

3 Understanding field of chemical engineering Solve problems related to atomic weight, is an official topic in Module 1 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding field of chemical engineering Solve problems related to atomic weight, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding field of chemical engineering solve problems related to atomic weight, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding field of chemical engineering Solve problems related to atomic weight, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding field of chemical engineering
Solve problems related to atomic weight, definition/meaning
steps, application. Technical terms English-

molecular weight and equivalent weight of a solid 4 Applying substance. Solve problems related to expressing

molecular weight and equivalent weight of a solid 4 Applying substance. Solve problems related to expressing is an official topic in Module 1 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify molecular weight and equivalent weight of a solid 4 Applying substance. Solve problems related to expressing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, molecular weight and equivalent weight of a solid 4 applying substance. solve problems related to expressing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What molecular weight and equivalent weight of a solid 4 Applying substance. Solve problems related to expressing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- $\square\square$ molecular weight and equivalent weight of a solid 4 Applying substance. Solve problems related to expressing $\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square$. $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square$, steps, application $\square\square\square\square$ $\square\square\square\square\square\square\square$ $\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square$.

4 Applying concentration of a solution and gases. Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems. Apply mechanism of heat transfer by conduction, convection, radiation to

4 Applying concentration of a solution and gases. Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems. Apply mechanism of heat transfer by conduction, convection, radiation to is an official topic in Module 1 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying concentration of a solution and gases. Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems. Apply mechanism of heat transfer by conduction, convection, radiation to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying concentration of a solution and gases. units and dimensions - various systems of units - fundamental & derived units- concept of unit

operations and unit process - gm atom, kg atom, gm mole, kg mole - atomic weight, molecular weight, equivalent weight - normality - molarity - molality - mass percent - mass, volume, mole relation, - relationships for gaseous systems - simple problems. apply mechanism of heat transfer by conduction, convection, radiation to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying concentration of a solution and gases. Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems. Apply mechanism of heat transfer by conduction, convection, radiation to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 Applying concentration of a solution and gases. Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems. Apply mechanism of heat transfer by conduction, convection, radiation to definition/meaning. steps, application. Technical terms English-.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

engineering.

M1.01	Explain the various systems of units	3
Understanding		

M1.02	Describe unit operations and unit process in the	3
Understanding		

M1.03	field of chemical engineering	
Applying	Solve problems related to atomic weight, molecular weight and equivalent weight of a solid	4

M1.04	substance.	
Applying	Solve problems related to expressing concentration of a solution and gases.	4

Contents:

Units and Dimensions - various systems of Units - Fundamental & Derived Units-
concept

of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic
weight,

Molecular weight, Equivalent Weight - normality - molarity - molality - mass
percent -

Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple
problems.

Module study routine: Read the source wording, make a one-page concept map,
practise one short answer and one structured answer, then review the Malayalam
support notes.

MODULE 2

solve problems in chemical engineering.

This module is presented from the official syllabus scope for Chemical Engineering.
Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: solve problems in chemical engineering.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to

Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify different modes of heat transfer 3 understanding apply fourier's law of thermal conduction to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓ ഓ Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to ഓ ഓ definition/meaning ഓ ഓ. ഓ ഓ ഓ, steps, application ഓ ഓ. Technical terms English-ഓ ഓ ഓ.

derive heat transfer equations for various solid 8 Applying surfaces and solve problems related to it.

derive heat transfer equations for various solid 8 Applying surfaces and solve problems related to it. is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify derive heat transfer equations for various solid 8 Applying surfaces and solve problems related to it. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, derive heat transfer equations for various solid 8 applying surfaces and solve problems related to it. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What derive heat transfer equations for various solid 8 Applying surfaces and solve problems related to it. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

Illustrate various heat exchangers used in industry

Illustrate various heat exchangers used in industry is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate various heat exchangers used in industry using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate various heat exchangers used in industry should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate various heat exchangers used in industry means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഈ Module-ൽ Illustrate various heat exchangers used in industry എന്നാണ് ഉദ്ദേശ്യം. definition/meaning എന്നാണ് ഉദ്ദേശ്യം. എന്നാണ് ഉദ്ദേശ്യം, steps, application എന്നാണ് ഉദ്ദേശ്യം. Technical terms English-ൽ എന്നാണ് ഉദ്ദേശ്യം.

Discuss the types of evaporators used in industry. 2 Understanding Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers - Evaporators - horizontal and vertical evaporators. Paraphrase mechanism of mass transfer by diffusion, absorption, leaching,

Discuss the types of evaporators used in industry. 2 Understanding Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers - Evaporators - horizontal and vertical evaporators. Paraphrase mechanism of mass transfer by diffusion, absorption, leaching, is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss the types of evaporators used in industry. 2 Understanding Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers - Evaporators - horizontal and vertical evaporators. Paraphrase mechanism of mass transfer by diffusion, absorption, leaching, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the types of evaporators used in industry. 2 understanding conduction - fourier's law - derivation - problems - conduction through plane wall, cylindrical wall - derivations - problems -, convection - concept of natural convection, radiation idea of black body - gray body - emissivity - absorptivity, heat exchanger patterns - shell and tube - double pipe heat exchangers - evaporators - horizontal and vertical evaporators. paraphrase mechanism of mass transfer by diffusion, absorption, leaching, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss the types of evaporators used in industry. 2 Understanding Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers - Evaporators - horizontal and vertical evaporators. Paraphrase mechanism of mass transfer by diffusion, absorption, leaching, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Discuss the types of evaporators used in industry. 2 Understanding Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers - Evaporators - horizontal and vertical evaporators. Paraphrase mechanism of mass transfer by diffusion, absorption, leaching, definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

solve problems in chemical engineering.

M2.01	Identify different modes of heat transfer	3
Understanding		
M2.02	Apply Fourier's law of thermal conduction to derive heat transfer equations for various solid	8
Applying		
	surfaces and solve problems related to it.	
M2.03	Illustrate various heat exchangers used in industry	2
Understanding		
M2.04	Discuss the types of evaporators used in industry.	2
Understanding		
	Series Test - I	1

Contents:

Conduction - Fourier's law - derivation - problems - Conduction through Plane wall,

Cylindrical wall - derivations - problems -, Convection - concept of natural convection,

Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger

patterns - Shell and Tube - Double Pipe Heat Exchangers – Evaporators - horizontal and

vertical evaporators.

Paraphrase mechanism of mass transfer by diffusion, absorption,

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

extraction, adsorption and drying.

This module is presented from the official syllabus scope for Chemical Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: extraction, adsorption and drying.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Understanding equipment's.

4 Understanding equipment's. is an official topic in Module 3 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding equipment's. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding equipment's. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding equipment's. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 4 Understanding equipment's. ഏതെങ്കിലും ഒരു ഉപകരണത്തിന്റെ definition/meaning എഴുതുക. ഉപയോഗം, ഏതെങ്കിലും ഏതെങ്കിലും, steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

Generalize leaching technique

Generalize leaching technique is an official topic in Module 3 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Generalize leaching technique using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, generalize leaching technique should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Generalize leaching technique means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ാം Generalize leaching technique ഏകീകരിക്കുന്നതിനുള്ള
രീതികൾ definition/meaning വ്യക്തമാക്കുന്നു. ഏകീകരിക്കാൻ ഏകീകരിക്കൽ, steps,
application ഏകീകരിക്കാൻ ഏകീകരിക്കൽ. Technical terms English-ൽ ഏകീകരിക്കൽ
ഏകീകരിക്കുന്നു.

Describe extraction and extraction equipment's

Describe extraction and extraction equipment's is an official topic in Module 3 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe extraction and extraction equipment's using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe extraction and extraction equipment's should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe extraction and extraction equipment's means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എറു Describe extraction and extraction equipment's ഏറു definition/meaning ഏറു. ഏറു ഏറു ഏറു, steps, application ഏറു ഏറു. Technical terms English-എറു ഏറു ഏറു.

Summarize drying equipment's used in industry 4 Understanding Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier

Summarize drying equipment's used in industry 4 Understanding Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier is an official topic in Module 3 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize drying equipment's used in industry 4 Understanding Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize drying equipment's used in industry 4 understanding diffusion - fick's law - law of conservation of mass - henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of bollman extractor - extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - drying - applications - working of tray

and spray drier should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize drying equipment's used in industry 4 Understanding Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Summarize drying equipment's used in industry 4 Understanding Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier definition/meaning . . . , steps, application Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

extraction, adsorption and drying.

Discuss gas absorption and gas absorption

M3.01

4

Understanding

equipment's.

M3.02

Generalize leaching technique

3

Understanding

M3.03

Describe extraction and extraction equipment's

3

Understanding

M3.04

Summarize drying equipment's used in industry

4

Understanding

Contents:

Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption

-

working of packed tower - flooding - channeling - leaching - applications - working of

Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption -

physical and chemical adsorption - Drying - applications - working of tray and spray drier

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Generalize the basic principles and types of distillation

This module is presented from the official syllabus scope for Chemical Engineering.
Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Generalize the basic principles and types of distillation
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive

Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive is an official topic in Module 4 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, differentiate various types of distillation. 3 understanding apply raoul's law and daltons law to derive should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓർമ്മപ്പെടുത്തൽ Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive ഓർമ്മപ്പെടുത്തൽ ഓർമ്മപ്പെടുത്തൽ definition/meaning ഓർമ്മപ്പെടുത്തൽ. ഓർമ്മപ്പെടുത്തൽ ഓർമ്മപ്പെടുത്തൽ, steps, application ഓർമ്മപ്പെടുത്തൽ ഓർമ്മപ്പെടുത്തൽ ഓർമ്മപ്പെടുത്തൽ. Technical terms English-ഓർമ്മപ്പെടുത്തൽ ഓർമ്മപ്പെടുത്തൽ.

equation connecting mole fraction and vapor of a 5 Applying binary mixtures

equation connecting mole fraction and vapor of a 5 Applying binary mixtures is an official topic in Module 4 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify equation connecting mole fraction and vapor of a 5 Applying binary mixtures using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, equation connecting mole fraction and vapor of a 5 applying binary mixtures should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What equation connecting mole fraction and vapor of a 5 Applying binary mixtures means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4- Δ equation connecting mole fraction and vapor of a 5 Applying binary mixtures Δ definition/meaning Δ . Δ Δ Δ , steps, application Δ Δ Δ . Technical terms English- Δ Δ Δ .

Derive equation for relative volatility

Derive equation for relative volatility is an official topic in Module 4 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Derive equation for relative volatility using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, derive equation for relative volatility should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Derive equation for relative volatility means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Derive equation for relative volatility

日本語の定義/意味を説明してください。 手順、適用、技術用語の英語-日本語の定義を説明してください。

Compare Azeotropic and Extractive Distillation 3 Understanding Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation. Text / Reference: T/R Book Title/Author Process Calculations for chemical engineers - Chem. Engg., Edn Development T1 Centre, IIT Madras R1 Process Calculations - V. Venkataramani, N. Anantharaman Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill R2 Inc. US; International 1955 Edition) R3 Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition) Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd R4 edition) Online Resources: Sl.No Website Link 1 <https://www.researchgate.net> 2 <http://cc.sit.edu.cn> 3 <http://www.chem.ucalgar>

Compare Azeotropic and Extractive Distillation 3 Understanding Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation. Text / Reference: T/R Book Title/Author Process Calculations for chemical engineers - Chem. Engg., Edn Development T1 Centre, IIT Madras R1 Process Calculations - V. Venkataramani, N. Anantharaman Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill R2 Inc. US; International 1955 Edition) R3 Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition) Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd R4 edition) Online Resources: Sl.No Website Link 1 <https://www.researchgate.net> 2 <http://cc.sit.edu.cn> 3 <http://www.chem.ucalgar> is an official topic in Module 4 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compare Azeotropic and Extractive Distillation 3 Understanding Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation. Text / Reference: T/R Book Title/Author Process Calculations for chemical engineers

- Chem. Engg., Edn Development T1 Centre, IIT Madras R1 Process Calculations - V. Venkataramani, N. Anantharaman Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill R2 Inc. US; International 1955 Edition) R3 Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition) Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd R4 edition) Online Resources: Sl.No Website Link 1 <https://www.researchgate.net> 2 <http://cc.sit.edu.cn> 3 <http://www.chem.ucalgary> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compare azeotropic and extractive distillation 3 understanding distillation - types - simple and steam distillation -fractional distillation - raoult's law - daltons law - relative volatility - derivation - simple problems - minimum and maximum boiling azeotropes -comparison - azeotropic and extractive distillation. text / reference: t/r book title/author process calculations for chemical engineers - chem. engg., edn development t1 centre, iit madras r1 process calculations - v. venkataramani, n. anantharaman introduction to chemical engineering - walter.l.badger (publisher: mcgraw-hill r2 inc. us; international 1955 edition) r3 unit operations - mccabe smith (publisher - mcgraw-hill education; 7edition) mass transfer - robert e treybal (publisher - mcgraw-hill book company; 3rd r4 edition) online resources: sl.no website link 1 <https://www.researchgate.net> 2 <http://cc.sit.edu.cn> 3 <http://www.chem.ucalgary> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compare Azeotropic and Extractive Distillation 3 Understanding Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation. Text / Reference: T/R Book Title/Author Process Calculations for chemical engineers - Chem. Engg., Edn Development T1 Centre, IIT Madras R1 Process Calculations - V. Venkataramani, N. Anantharaman Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill R2 Inc. US; International 1955 Edition) R3 Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition) Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd R4 edition) Online Resources: Sl.No Website Link 1 https://www.researchgate.net 2 http://cc.sit.edu.cn 3 http://www.chem.ucalgary means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-Compare Azeotropic and Extractive Distillation
3 Understanding Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation. Text / Reference: T/R Book Title/Author Process Calculations for chemical engineers - Chem. Engg., Edn Development T1 Centre, IIT Madras R1 Process Calculations - V. Venkataramani, N. Anantharaman Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill R2 Inc. US; International 1955 Edition) R3 Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition) Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd R4 edition) Online Resources: Sl.No Website Link 1 <https://www.researchgate.net> 2 <http://cc.sit.edu.cn> 3 <http://www.chem.ucalgary>
പ്രതിരോധം പ്രതിരോധം definition/meaning പ്രതിരോധം പ്രതിരോധം, steps, application പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം. Technical terms English-പ്രതിരോധം പ്രതിരോധം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Generalize the basic principles and types of distillation

M4.01	Differentiate various types of distillation.	3
Understanding		
M4.02	Apply Raoul's law and Daltons law to derive equation connecting mole fraction and vapor of a	5
Applying	binary mixtures	
M4.03	Derive equation for relative volatility	4
Applying		
M4.04	Compare Azeotropic and Extractive Distillation	3
Understanding		
	Series Test - II	1

Contents:

Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation.

Text / Reference:

T/R	Book Title/Author
Development	Process Calculations for chemical engineers - Chem. Engg., Edn
T1	

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Explain the various systems of units 3 Understanding Describe unit operations and unit process in the. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the various systems of units 3 Understanding Describe unit operations and unit process in the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding field of chemical engineering Solve problems related to atomic weight,. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding field of chemical engineering Solve problems related to atomic weight,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain molecular weight and equivalent weight of a solid 4 Applying substance. Solve problems related to expressing. (3 marks)

Answer: Begin with the standard definition or identity of *molecular weight and equivalent weight of a solid 4 Applying substance. Solve problems related to expressing.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Applying concentration of a solution and gases. Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems. Apply mechanism of heat transfer by conduction, convection, radiation to. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying concentration of a solution and gases. Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm*

mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems. Apply mechanism of heat transfer by conduction, convection, radiation to. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 2

Define or explain Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to. (3 marks)

Answer: Begin with the standard definition or identity of *Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain derive heat transfer equations for various solid 8 Applying surfaces and solve problems related to it.. (3 marks)

Answer: Begin with the standard definition or identity of *derive heat transfer equations for various solid 8 Applying surfaces and solve problems related to it..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Illustrate various heat exchangers used in industry. (3

marks)

Answer: Begin with the standard definition or identity of *Illustrate various heat exchangers used in industry*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Discuss the types of evaporators used in industry. 2 Understanding Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers - Evaporators - horizontal and vertical evaporators. Paraphrase mechanism of mass transfer by diffusion, absorption, leaching,. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss the types of evaporators used in industry. 2 Understanding Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers - Evaporators - horizontal and vertical evaporators. Paraphrase mechanism of mass transfer by diffusion, absorption, leaching,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain 4 Understanding equipment's.. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding equipment's..* State the governing principle, list the key components or stages, and close with one

relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Generalize leaching technique. (3 marks)

Answer: Begin with the standard definition or identity of *Generalize leaching technique*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Describe extraction and extraction equipment's. (3 marks)

Answer: Begin with the standard definition or identity of *Describe extraction and extraction equipment's*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Summarize drying equipment's used in industry 4 Understanding Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize drying equipment's used in industry 4 Understanding Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier*.

raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive. (3 marks)

Answer: Begin with the standard definition or identity of *Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain equation connecting mole fraction and vapor of a 5 Applying binary mixtures. (3 marks)

Answer: Begin with the standard definition or identity of *equation connecting mole fraction and vapor of a 5 Applying binary mixtures*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Derive equation for relative volatility. (3 marks)

Answer: Begin with the standard definition or identity of *Derive equation for relative volatility*. State the governing principle, list the key components or stages, and close

with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Compare Azeotropic and Extractive Distillation 3

Understanding Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation. Text / Reference: T/R Book Title/Author Process Calculations for chemical engineers - Chem. Engg., Edn Development T1 Centre, IIT Madras R1 Process Calculations - V. Venkataramani, N. Anantharaman Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill R2 Inc. US; International 1955 Edition) R3 Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition) Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd R4 edition) Online Resources: Sl.No Website Link 1 <https://www.researchgate.net> 2 <http://cc.sit.edu.cn> 3 <http://www.chem.ucalgar>. (3 marks)

Answer: Begin with the standard definition or identity of *Compare Azeotropic and Extractive Distillation 3 Understanding Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation. Text / Reference: T/R Book Title/Author Process Calculations for chemical engineers - Chem. Engg., Edn Development T1 Centre, IIT Madras R1 Process Calculations - V. Venkataramani, N. Anantharaman Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill R2 Inc. US; International 1955 Edition) R3 Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition) Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd R4 edition) Online Resources: Sl.No Website Link 1 <https://www.researchgate.net> 2 <http://cc.sit.edu.cn> 3 <http://www.chem.ucalgar>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain the various systems of units 3 Understanding Describe unit operations and unit process in the.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **4 Understanding equipment's..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.

Term	Meaning in this lesson
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.researchgate.net> <http://cc.sit.edu.cn>
<http://www.chem.ucalgary>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5093C. Verify institutional updates against the current official curriculum.